

A. Customer Journey

customer_id integer	plan_id integer	start_date date	plan_id integer	plan_name character varying (13)	price numeric (5,2)
1	0	2020-08-01	0	trial	0.00
1	1	2020-08-08	1	basic monthly	9.90

The customer_id = 1 went from the trial plan to basic monthly plan instead of going to the default pro monthly plan

2	0	2020-09-20	0	trial	0.00
2	3	2020-09-27	3	pro annual	199.00

The customer_id = 2 went from the trial plan to pro annual plan instead of going to the default pro monthly plan

3	0	2020-01-13	0	trial	0.00
3	1	2020-01-20	1	basic monthly	9.90

The customer_id = 3 went from the trial plan to basic monthly plan instead of going to the default pro monthly plan

4	0	2020-01-17	0	trial	0.00
4	1	2020-01-24	1	basic monthly	9.90
4	4	2020-04-21	4	churn	[null]

The customer_id = 4 went from trial plan to basic monthly plan and then cancelled the plan on 2020-04-21

5	0	2020-08-03	0	trial	0.00
5	1	2020-08-10	1	basic monthly	9.90

The customer_id = 5 went from the trial plan to basic monthly plan instead of going to the default pro monthly plan

6	0	2020-12-23	0	trial	0.00
6	1	2020-12-30	1	basic monthly	9.90
6	4	2021-02-26	4	churn	[null]

The customer_id = 6 went from trial plan to basic monthly plan and then cancelled the plan on 2021-02-26

7	0	2020-02-05	0	trial	0.00
7	1	2020-02-12	1	basic monthly	9.90
7	2	2020-05-22	2	pro monthly	19.90

The customer_id = 7 had a history of sequential upgrades from trial to basic monthly to pro monthly, this customer is having high potential of becoming a pro annual subscriber

8	0	2020-06-11	0	trial	0.00
8	1	2020-06-18	1	basic monthly	9.90
8	2	2020-08-03	2	pro monthly	19.90

The customer_id = 8 has the same history of upgrades like customer_id = 7

B. Data Analysis Questions

--1. How many customers has Foodie-Fi ever had?

```
select count(DISTINCT customer_id) as total_customers
from subscriptions;
```

	total_customers bigint 
1	1000

--2. What is the monthly distribution of trial plan start_date values
-- for our dataset - use the start of the month as the group by value

```
with month_tbl as (  
  select *, EXTRACT(month from start_date) as month  
  from subscriptions  
  where plan_id = 0  
)  
select month, count(*) as count_of_plans  
from month_tbl  
group by month  
order by month;
```

	month numeric 🔒	count_of_plans bigint 🔒
1	1	88
2	2	68
3	3	94
4	4	81
5	5	88
6	6	79
7	7	89
8	8	88
9	9	87
10	10	79
11	11	75
12	12	84

--3. What plan start_date values occur after the year 2020 for our
-- dataset? Show the breakdown by count of events for each plan_name

```
with cnt as (  
    select plan_id, count(*) as no_of_plans  
    from subscriptions  
    where EXTRACT(year from start_date) > 2020  
    group by plan_id  
)  
select plan_name, no_of_plans  
from cnt  
left join plans p  
on cnt.plan_id = p.plan_id  
order by no_of_plans DESC;
```

	plan_name character varying (13) 🔒	no_of_plans bigint 🔒
1	churn	71
2	pro annual	63
3	pro monthly	60
4	basic monthly	8

--4. What is the customer count and percentage of customers
-- who have churned rounded to 1 decimal place?

```
with plan_cnt as (  
    select plan_name, count(*) as no_of_plans  
    from subscriptions s  
    left join plans p  
    on s.plan_id = p.plan_id  
    group by plan_name  
)  
select plan_name, no_of_plans, ROUND(100 * no_of_plans::decimal / total_plans, 1) as percentage  
from plan_cnt pc  
CROSS JOIN (  
    SELECT COUNT(DISTINCT customer_id) as total_plans  
    FROM subscriptions  
)  
where plan_name = 'churn';
```

	plan_name character varying (13) 🔒	no_of_plans bigint 🔒	percentage numeric 🔒
1	churn	307	30.7

```
--5. How many customers have churned straight after their initial
-- free trial - what percentage is this rounded to the nearest whole number?
```

```
with sub_rnum as (
  select *, ROW_NUMBER() OVER(
    PARTITION BY customer_id
    ORDER BY start_date
  ) as r_num
  from subscriptions
)
, cnt as (
  select count(DISTINCT customer_id) as count_of_customers
  from sub_rnum
  where(r_num = 2 and plan_id = 4)
)
select count_of_customers, ROUND((100 * count_of_customers::decimal/total_plans)) as percentage_of_trial_to_churn
from cnt
CROSS JOIN (
  SELECT COUNT(DISTINCT customer_id) as total_plans
  FROM subscriptions
)
```

	count_of_customers bigint	percentage_of_trial_to_churn numeric
1	92	9

```
--6. What is the number and percentage of customer plans after their initial free trial?
```

```
with next_pln_tbl as (
  select customer_id, plan_id,
    LEAD(plan_id) OVER(
      PARTITION BY customer_id
      ORDER BY plan_id
    ) as next_plan_id
  from subscriptions
)
select next_plan_id, count(*) as next_plan_count,
  ROUND(
    100*count(*)::decimal/(SELECT COUNT(DISTINCT customer_id) as total_plans FROM subscriptions)
    ,2) as next_plan_percentage
from next_pln_tbl
where plan_id = 0 and next_plan_id is not null
group by next_plan_id;
```

	next_plan_id integer	next_plan_count bigint	next_plan_percentage numeric
1	3	37	3.70
2	4	92	9.20
3	2	325	32.50
4	1	546	54.60

--7. What is the customer count and percentage breakdown of all 5 plan_name values at 2020-12-31?

```
select * from plans;
select * from subscriptions;

with next_pln_tbl as (
  select *, LEAD(plan_id) OVER(PARTITION BY customer_id ORDER BY start_date) as next_plan_id
  from subscriptions
  where start_date <= '2020-12-31'
)
select plan_id, count(*) as plan_count,
  ROUND(
    100*count(*)::decimal/(select count(distinct customer_id) from subscriptions where start_date <= '2020-12-31')
    ,2) as plan_percentage
from next_pln_tbl
where next_plan_id is null
group by plan_id;
```

	plan_id integer	plan_count bigint	plan_percentage numeric
1	0	19	1.90
2	1	224	22.40
3	2	326	32.60
4	3	195	19.50
5	4	236	23.60

--8. How many customers have upgraded to an annual plan in 2020?

```
select count(distinct customer_id) as no_of_customers_upgraded_to_annual_plan
from subscriptions
where EXTRACT(year from start_date) = 2020 and plan_id = 3;
```

	no_of_customers_upgraded_to_annual_plan bigint
1	195

--9. How many days on average does it take for a customer
-- to an annual plan from the day they join Foodie-Fi?

```
with days_tbl as (
  select *, (s2.start_date - s1.start_date) as days_to_transfer_to_annual
  from subscriptions s1
  join subscriptions s2
  on s1.customer_id = s2.customer_id
  where s1.plan_id = 0 and s2.plan_id = 3
)
select ROUND(AVG(days_to_transfer_to_annual)) as avg_days_it_took_customer_to_transfer_to_annual_plan
from days_tbl
```

	avg_days_it_took_customer_to_transfer_to_annual_plan numeric
1	105

```
--10. Can you further breakdown this average value
--into 30 day periods (i.e. 0-30 days, 31-60 days etc)
```

```
with days_tbl as (
    select *,(s2.start_date - s1.start_date) as days_to_transfer_to_annual
    from subscriptions s1
    join subscriptions s2
    on s1.customer_id = s2.customer_id
    where s1.plan_id = 0 and s2.plan_id = 3
),
bins_tbl as (
    select *,
        WIDTH_BUCKET(days_to_transfer_to_annual,0,365,12) as bins
    from days_tbl
),
bin_nm as (
    select *, ((bins - 1)*30 || ' - ' || bins * 30 || ' days') as bucket
    from bins_tbl
)
select bins,bucket,COUNT(*) as count_of_customers
from bin_nm
group by bins,bucket
order by bins
```

	bins integer	bucket text	count_of_customers bigint
1	1	0 - 30 days	49
2	2	30 - 60 days	24
3	3	60 - 90 days	35
4	4	90 - 120 days	35
5	5	120 - 150 days	43
6	6	150 - 180 days	37
7	7	180 - 210 days	24
8	8	210 - 240 days	4
9	9	240 - 270 days	4
10	10	270 - 300 days	1
11	11	300 - 330 days	1
12	12	330 - 360 days	1

--11. How many customers downgraded from a pro monthly to a basic monthly plan in 2020?

```
select *  
from subscriptions s1  
join subscriptions s2  
on s1.customer_id = s2.customer_id  
where s1.plan_id = 3 and s2.plan_id = 1 and s1.start_date < s2.start_date;
```

customer_id	plan_id	start_date	customer_id	plan_id	start_date
integer	integer	date	integer	integer	date

No result cuz there wasn't any customer who downgraded from pro monthly to basic monthly plan i guess they must be really liking the pro monthly :D

C. Challenge Payment Question